

HELP DOCUMENTATION

[API](#)

[How To](#)

[FAQ](#)

[Legend](#)

[TLE Format](#)

[Data Requests](#)

[Laser Clearinghouse](#)

[SGP4](#)

[User Agreement](#)

[Privacy](#)

[Contact Us](#)

Basic Description of the Two Line Element (TLE) Format

[Basic Info](#)

[Additional Info](#)

[Alpha 5](#)

[Alpha 5 Converter](#)



(<https://www.facebook.com/SpaceTrack>)



(<https://twitter.com/spacetrackorg>)

Developed by SAIC

(<http://www.saic.com>) under contract to S4S-CJFSCC. Contact Us (<https://www.space-track.org/documentation#/contact>)

[↑ Back to top](#)

Basic Info

ISS (ZARYA)

1 25544U 98067A 04236.56031392 .00020137 00000-0 16538-3 0 9993
 2 25544 51.6335 344.7760 0007976 126.2523 325.9359 15.70406856328906

 123456789012345678901234567890123456789012345678901234567890 reference number
 1 2 3 4 5 6 7

Line 0

Columns	Example	Description
1-24	ISS (ZARYA)	The common name for the object based on information from the Satellite Catalog.

Line 1

Columns	Example	Description
1	1	Line Number
3-7	25544	Satellite Catalog Number
8	U	Elset Classification
10-17	98067A	International Designator
19-32	04236.56031392	Element Set Epoch (UTC) *Note: spaces are acceptable in columns 21 & 22
34-43	.00020137	1st Derivative of the Mean Motion with respect to Time
45-52	00000-0	2nd Derivative of the Mean Motion with respect to Time (decimal point assumed)
54-61	16538-3	B* Drag Term
63	0	Element Set Type
65-68	999	Element Number

Line 0		
Columns	Example	Description
Line 2		
Columns	Example	Description
1	2	Line Number
3-7	25544	Satellite Catalog Number
9-16	51.6335	Orbit Inclination (degrees)
18-25	344.7760	Right Ascension of Ascending Node (degrees)
27-33	0007976	Eccentricity (decimal point assumed)
35-42	126.2523	Argument of Perigee (degrees)
44-51	325.9359	Mean Anomaly (degrees)
53-63	15.70406856	Mean Motion (revolutions/day)
64-68	32890	Revolution Number at Epoch
69	6	Checksum

↑ Back to top

Additional Information

Satellite Catalog Number

example: 25544

- The catalog number assigned to the object by the US Space Force.
- Numbers are assigned sequentially as objects are cataloged.
- Object numbers less than 10000 are always aligned to the right, and padded with zeros or spaces to the left.



(<https://www.facebook.com/SpaceTrack>)



(<https://twitter.com/spacetrackorg>)

Developed by SAIC

(<http://www.saic.com>) under contract to S4S-CJFSCC. Contact Us (<https://www.space-track.org/documentation#/contact>)

↑ Back
to top

The security classification of the element set. All objects on this site will have a classification of 'U' (unclassified).

International Designator

example: 98067A

This is another format for identifying an object.

- The first two characters designate the launch year of the object.
- The next 3 characters indicate the launch number, starting from the beginning of the year.
This particular launch was the 67th launch of 1998.
- The remainder of the field (1 to 3 characters) indicates the piece of the launch. Piece 'A' is usually the payload.

Element Set Epoch

example: 04236.56031392

- The first two digits ('04') indicate the year. Add 1900 for years ≥ 57 , and 2000 for all others.
- The remainder of the field ('236.56031392') is the day of the year.
- Spaces or numbers are acceptable in day of the year. (e.g. '236' or '006' or '6').

1st Derivative of the Mean Motion

example: .00020137

"+", "-", or " " (space) are acceptable in the 34th character position on Line 1, and indicate positive or negative value for 1st Derivative of the Mean Motion. A space implies a positive value.

Checksum

The checksum is the sum of all characters in the data line, modulo 10. In this formula, the following non-numeric characters are assigned the indicated values:

- Blanks, periods, letters, '+' signs -> 0
- '-' signs -> 1



(<https://www.facebook.com/SpaceTrack>)



(<https://twitter.com/spacetrackorg>)

Developed by SAIC

(<http://www.saic.com>) under contract to S4S-CJFSCC. Contact Us (<https://www.space-track.org/documentation#/contact>)

↑ Back to top

↑ Back to top

Alpha 5

Alpha-5

Alpha-5 is a stopgap object numbering schema from the United States Space Force that increases the satellite catalog’s capacity to display up to 339,999 objects in the GP/GP_History API classes using legacy fixed-width Two and Three Line Element Set (TLE/3LE) formats.

Replacing the 1st digit of the 5-digit object number with an alphanumeric character makes it possible to represent 240,000 more numbers. Objects less than 100,000 are unaffected by Alpha-5, as are users who download elsets from the GP and GP_History API classes in other formats like XML, JSON, KVN, and CSV. In order to preserve legacy operations that depend on 5-digit integers, our legacy API Classes tle, tle_latest, and tle_publish will not change to Alpha-5.

Only capital letters and numbers are used in Alpha-5. The letters “I” and “O” are omitted to avoid confusion with the numbers “1” and “0”.

Examples

Some example object numbers in Alpha-5 format, as they would be seen in the TLE:

100000 => A0000	148493 => E8493
182931 => J2931	234018 => P4018
301928 => W1928	339999 => Z9999

Even though TLE/3LE will support Alpha-5 format on output in lines 1 and 2, users need to specify NORAD_CAT_ID integer values to return specific objects. The API will not support filtering on Alpha-5 values or ranges like /NORAD_CAT_ID/T0000--V9999/.

Alpha-5 Propagator

If your propagation software cannot process alphabetical characters in the catalog number section of the TLE, you can eliminate the Alpha-5 range from your query by adding /NORAD_CAT_ID/<100000/ to filter them out as displayed in this example: https://www.space-track.org/basicspace-data/query/class/NORAD_CAT_ID/%3C100000/development/val/epoch/%3Enow

 [facebook.com/space-track.org](https://www.facebook.com/space-track.org)

 twitter.com/space-track-org

 [youtube.com/space-track-org](https://www.youtube.com/space-track-org)

 [linkedin.com/company/space-track-org](https://www.linkedin.com/company/space-track-org)

(http://www.saic.com) under contract to S4S-CJFSCC. Contact Us (<https://www.space-track.org/documentation#/contact>)

Back to top

30/orderby/norad_cat_id/format/tle (https://www.space-track.org/basicspacedata/query/class/gp/NORAD_CAT_ID/%3C100000/decay_date/null-val/epoch/%3Enow-30/orderby/norad_cat_id/format/tle)

The latest SGP4 propagator, Version 8.0, is compatible with the new Alpha-5 format shown in this graphic and is available at <https://www.space-track.org/documentation#/sgp4> (<https://www.space-track.org/documentation#/sgp4>):

0										1										2										3										4										5										6																			
1 2 3 4 5 6 7 8 9										0 1 2 3 4 5 6 7 8 9										0 1 2 3 4 5 6 7 8 9										0 1 2 3 4 5 6 7 8 9										0 1 2 3 4 5 6 7 8 9										0 1 2 3 4 5 6 7 8 9										0 1 2 3 4 5 6 7 8 9																			
#0 Legacy TLE																																																																															
1 2 1 2 3 3 U										9 8 0 6 7 A										0 4 2 3 6 . 5 6 0 3 1 3 9 2										. 0 0 0 2 0 1 3 7										0 0 0 0 0 - 0										1 6 5 3 8 - 3										0 9 9 9 9 3																			
2 2 1 2 3 3										5 1 . 6 3 3 5										3 4 4 . 7 7 6 0										0 0 0 7 9 7 6										1 2 6 . 2 5 2 3										3 2 5 . 9 3 5 9										1 5 . 7 0 4 0										6 8 5 6 3 2 8 9 0 3									
#1 Alpha-5																																																																															
1 A 5 5 4 4 U										9 8 0 6 7 A										0 4 2 3 6 . 5 6 0 3 1 3 9 2										. 0 0 0 2 0 1 3 7										0 0 0 0 0 - 0										1 6 5 3 8 - 3										0 9 9 9 9 3																			
2 A 5 5 4 4										5 1 . 6 3 3 5										3 4 4 . 7 7 6 0										0 0 0 7 9 7 6										1 2 6 . 2 5 2 3										3 2 5 . 9 3 5 9										1 5 . 7 0 4 0										6 8 5 6 3 2 8 9 0 3									
Legend																																																																															
Satellite #										Int'l designator										Epoch Time										N-Dot/2										N-DoubleDot/6										Bstar										ELSET #																			
Satellite #										Inclination										Right Ascension										Eccentricity										Arg of Perigee										Mean Anomaly										Mean Motion										Epoch Rev									

Full Alpha-5 Numbering

A => 10	J => 18	S => 26
B => 11	K => 19	T => 27
C => 12	L => 20	U => 28
D => 13	M => 21	V => 29
E => 14	N => 22	W => 30
F => 15	P => 23	X => 31
G => 16	Q => 24	Y => 32
H => 17	R => 25	Z => 33

Future Use

While this schema extends the usability of the TLE format, space-track.org encourages users to switch to a more extensible format, such as XML, KVN, or JSON. These formats handle the ever-increasing size of the catalog, regardless of object number size.



Back to top

(<https://www.facebook.com/SpaceTrack>)



(<https://twitter.com/spacetrackorg>)

Developed by SAIC

(<http://www.saic.com>) under contract to S4S-CJFSCC. Contact Us (<https://www.space-track.org/documentation#/contact>)

↑ Back to top

Alpha-5 Converter

[↑ Back to top](#)

